

# Day 7: Tuples, Sets & Dictionaries in Python (Detailed Notes for Data Analysis Students)

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## Introduction

So far you've learned:

- Variables
- Operators
- Conditional Statements
- Loops
- Strings
- Lists

Today you'll learn three important data structures:

1. Tuple
2. Set
3. Dictionary

These are heavily used in:

- Data Analysis
  - Pandas
  - SQL Integration
  - APIs
  - Machine Learning
- 

## 1. TUPLES

### WHAT IS A TUPLE?

A tuple is a collection of items that is:

- ✓ Ordered
  - ✓ Allows Duplicates
  - ✗ Cannot Be Modified (Immutable)
- 

### CREATING A TUPLE

```
employees = ("Rahul", "Amit", "Priya")

print(employees)
```

Output:

```
('Rahul', 'Amit', 'Priya')
```

---

## WHY USE TUPLE?

Suppose company departments never change.

```
departments = (
    "HR",
    "Sales",
    "Finance",
    "IT"
)
```

Using tuple prevents accidental modifications.

---

## ACCESS TUPLE VALUES

```
employees = ("Rahul","Amit","Priya")

print(employees[0])
```

Output:

```
Rahul
```

---

## NEGATIVE INDEXING

```
employees = ("Rahul","Amit","Priya")

print(employees[-1])
```

Output:

```
Priya
```

---

## TUPLE LENGTH

```
employees = ("Rahul","Amit","Priya")

print(len(employees))
```

Output:

```
3
```

---

## TUPLE CANNOT BE CHANGED

```
employees = ("Rahul","Amit")

employees[0] = "Vikas"
```

Output:

```
TypeError
```

---

## LOOP THROUGH TUPLE

```
employees = ("Rahul","Amit","Priya")

for emp in employees:
    print(emp)
```

# Tuple Functions

## COUNT()

```
numbers = (10,20,10,30)

print(numbers.count(10))
```

Output:

```
2
```

---

## INDEX()

```
numbers = (10,20,30)

print(numbers.index(20))
```

Output:

```
1
```

---

## Real Data Analysis Example

```
months = (  
    "Jan",  
    "Feb",  
    "Mar",  
    "Apr"  
)  
  
for month in months:  
    print(month)
```

---

## 2. SETS

### WHAT IS A SET?

A set is a collection that:

-  No Duplicates
  -  No Indexing
  -  No Ordering
  -  Fast Searching
- 

### CREATE A SET

```
cities = {  
    "Mumbai",  
    "Delhi",  
    "Pune"  
}  
  
print(cities)
```

---

### DUPLICATE VALUES REMOVED

```
cities = {  
    "Mumbai",
```

```
"Delhi",  
"Mumbai"  
}  
  
print(cities)
```

Output:

```
{'Mumbai', 'Delhi'}
```

---

## WHY SETS ARE USEFUL?

Suppose customer data contains duplicates.

```
customers = [  
    "Rahul",  
    "Amit",  
    "Rahul",  
    "Priya"  
]
```

Convert to set:

```
customers = set(customers)  
  
print(customers)
```

Output:

```
{'Rahul', 'Amit', 'Priya'}
```

Duplicates removed automatically.

---

## Add Item

```
cities = {"Mumbai", "Pune"}  
  
cities.add("Delhi")  
  
print(cities)
```

---

## Remove Item

```
cities = {"Mumbai","Pune"}

cities.remove("Pune")

print(cities)
```

---

## Loop Through Set

```
cities = {"Mumbai","Delhi","Pune"}

for city in cities:
    print(city)
```

---

## Set Operations

### UNION

```
A = {1,2,3}
B = {3,4,5}

print(A.union(B))
```

Output:

```
{1,2,3,4,5}
```

---

### INTERSECTION

Common values.

```
A = {1,2,3}
B = {2,3,4}

print(A.intersection(B))
```

Output:

```
{2,3}
```

---

### DIFFERENCE

```
A = {1,2,3}
B = {2,3,4}

print(A.difference(B))
```

Output:

```
{1}
```

---

## Real Data Analyst Example

Find common customers:

```
branch1 = {
    "Rahul",
    "Amit",
    "Priya"
}

branch2 = {
    "Priya",
    "Amit",
    "Neha"
}

print(
    branch1.intersection(branch2)
)
```

Output:

```
{'Amit', 'Priya'}
```

---

## 3. DICTIONARIES

### WHAT IS A DICTIONARY?

Dictionary stores data as:

```
Key : Value
```

Example:

Employee ID : Employee Name

---

## Create Dictionary

```
employee = {  
    "Name": "Rahul",  
    "Salary": 50000,  
    "Department": "Sales"  
}  
  
print(employee)
```

Output:

```
{  
  'Name': 'Rahul',  
  'Salary': 50000,  
  'Department': 'Sales'  
}
```

---

## Why Dictionaries?

Instead of:

```
name = "Rahul"  
salary = 50000  
department = "Sales"
```

Store everything together:

```
employee = {  
    "Name": "Rahul",  
    "Salary": 50000,  
    "Department": "Sales"  
}
```

---

## Access Values

```
employee = {  
    "Name": "Rahul",
```



```
        "Salary":50000
    }

    print(employee["Name"])
```

Output:

```
Rahul
```

---

## Using get()

```
employee = {
    "Name": "Rahul"
}

print(employee.get("Name"))
```

Output:

```
Rahul
```

---

## Change Value

```
employee = {
    "Name": "Rahul"
}

employee["Name"] = "Amit"

print(employee)
```

---

## Add New Value

```
employee = {
    "Name": "Rahul"
}

employee["Salary"] = 50000

print(employee)
```

---

## Delete Item

```
employee = {  
    "Name": "Rahul",  
    "Salary": 50000  
}  
  
del employee["Salary"]  
  
print(employee)
```

---

## Dictionary Methods

### KEYS()

```
employee = {  
    "Name": "Rahul",  
    "Salary": 50000  
}  
  
print(employee.keys())
```

---

### VALUES()

```
print(employee.values())
```

---

### ITEMS()

```
print(employee.items())
```

---

## Loop Through Dictionary

```
employee = {  
    "Name": "Rahul",  
    "Salary": 50000,  
    "Department": "Sales"  
}  
  
for key, value in employee.items():
```

```
print(key, value)
```

Output:

```
Name Rahul  
Salary 50000  
Department Sales
```

---

## Nested Dictionary

```
employees = {  
    101:{  
        "Name":"Rahul",  
        "Salary":50000  
    },  
  
    102:{  
        "Name":"Amit",  
        "Salary":60000  
    }  
}  
  
print(employees[101]["Name"])
```

Output:

```
Rahul
```

---

## Real Data Analyst Example

Employee Database

```
employees = {  
    101:"Rahul",  
    102:"Amit",  
    103:"Priya"  
}  
  
for empid, name in employees.items():
```

```
print(empid, name)
```

---

## Mini Project 1: Employee Information

```
employee = {  
    "Name": "Rahul",  
    "Salary": 50000,  
    "Department": "Sales"  
}  
  
for k,v in employee.items():  
  
    print(k,":",v)
```

---

## Mini Project 2: Remove Duplicate Cities

```
cities = [  
    "Mumbai",  
    "Delhi",  
    "Mumbai",  
    "Pune"  
]  
  
cities = set(cities)  
  
print(cities)
```

---

## Mini Project 3: Monthly Sales

```
sales = {  
    "Jan": 50000,  
    "Feb": 60000,  
    "Mar": 70000  
}  
  
print(sum(sales.values()))
```

Output:

## Practice Questions

### TUPLES

1. Create a tuple of cities.
  2. Create a tuple of months.
  3. Print first item.
  4. Print last item.
  5. Count duplicate values.
  6. Find index of item.
  7. Loop through tuple.
  8. Find tuple length.
  9. Create salary tuple.
  10. Print all salaries.
- 

### SETS

11. Create a set.
  12. Add city.
  13. Remove city.
  14. Remove duplicates.
  15. Perform union.
  16. Perform intersection.
  17. Perform difference.
  18. Check item exists.
  19. Count unique values.
  20. Create customer set.
- 

### DICTIONARIES

21. Create employee dictionary.
22. Print employee name.
23. Print salary.
24. Update salary.
25. Add department.
26. Delete key.

27. Print keys.
  28. Print values.
  29. Print items.
  30. Loop through dictionary.
- 

## BUSINESS PROBLEMS

31. Employee database.
  32. Product inventory.
  33. Monthly sales report.
  34. Customer records.
  35. Department records.
  36. Remove duplicate customers.
  37. Find common customers.
  38. Calculate yearly sales.
  39. Generate employee report.
  40. Create company directory.
- 

## Assignment (Data Analyst Level)

Create a Company Employee System:

```
employees = {  
    101:{  
        "Name":"Rahul",  
        "Salary":50000  
    },  
    102:{  
        "Name":"Amit",  
        "Salary":60000  
    },  
    103:{  
        "Name":"Priya",  
        "Salary":55000  
    }  
}
```

Display:

```
Employee ID : 101
Name       : Rahul
Salary     : 50000
```

```
Employee ID : 102
Name       : Amit
Salary     : 60000
```

Also calculate:

- Total Salary
  - Average Salary
  - Highest Salary
  - Lowest Salary
- 

## Interview Questions

1. What is a tuple?
2. Difference between tuple and list?
3. What is a set?
4. Why are sets used?
5. What is a dictionary?
6. Difference between list and dictionary?
7. What is a key-value pair?
8. What is a nested dictionary?
9. How are dictionaries used in data analysis?
10. Give a real-world example of a set.

## Learning Outcome

After Day 7, you should be able to:

- ✓ Use Tuples for fixed data
- ✓ Use Sets to remove duplicates
- ✓ Use Dictionaries for structured records
- ✓ Build employee and sales databases
- ✓ Store and analyze business data
- ✓ Prepare for Pandas DataFrame concepts